DCDC LLC + 4SBB G474 — Design Notes

Oct 17, 2026 · @Chotiwat Preamprasertsook

Summary

The converter is now isolated and has two stages. A fixed-ratio LLC (DCX, about 205 kHz) gives 3 kVrms basic isolation and an isolated bus roughly equal to VIN. The original 4SBB then regulates on the secondary side, still controlled by the STM32G474 HRTIM. Modelled efficiency at 50 W is 91.4-91.8% from 24-48 V in and 86.9% at 12 V in. Worst-case loss is 7.5 W, at 12 V in.

Parameter	Target	Design
Input	12-48 V DC	12-48 V, 51 V TVS, 7 A fuse (primary)
Output	1.5-48 V, 5 A, 50 W	CV/CC envelope: 5 A up to 10 V, 50 W above
Isolation	Basic, about 3 kVrms	PQ2620 transformer + CA-IS3742LW isolator (5 kVrms) + PA3855 flyback aux supply
Topology	LLC + regulator, cost premium at most 20%	LLC DCX 1:2:2 -> secondary 4SBB. Premium is about 55% (see BOM)
Controller	STM32G474RBT6	On the secondary. HRTIM C = LLC, D = SR, A/E = 4SBB
Cooling	Passive, 25-30 C	4.5-7.5 W total; hottest parts are the 4SBB FETs (up to 1.4 W)
Comms	Isolated CAN	CA-IS3062VW, 5 kVrms
Address	6-bit DIP + 6-pin header	DIP and header in parallel on PC0-PC5
CAD	EasyEDA Pro	KiCad 6-format project zip for import

Deliverables (Rev D): an importable project zip (9-sheet wired schematic + fully routed 160 x 105 mm 4-layer PCB), Gerbers / drill / CPL plus the assembly upload BOM and CPL for JLCPCB, schematic and layer PDFs, BOMs, and pin-map.xlsx. The netlist was checked pin by pin against the design: 134 nets, 257 parts, 0 mismatches. A second check confirmed that only T1, U12 and the optional Y-cap connect primary nets to secondary nets.

Rev D changes: NTC moved from PA4 to PA6, so PA4 is free for DAC1_OUT1. 10k pull-downs added on the unused MCU pins (R93 now doubles as the PA2 pull-down for the VIN monitor). HSE changed to 16 MHz. Primary input OVP (50.26 V rising) reports through U12 channel 1 to PA15 / HRTIM1_FLT2, and an isolated VIN monitor (ramp-comparator PWM) reports through channel 2 to PA2 / TIM15_CH1. The ADC, DAC and comparator reference is now an external REF3030 (3.000 V ±0.2 %) on VREF+, no longer the isolator's 3.3 V rail. Fifteen test pads added on quiet nodes. PQ2620 footprint pitch corrected to 3.8

mm within a group of pins and 7.5 mm between pins 3 and 4. Schematic text overlaps cleaned up. The PCB was re-placed and re-routed with tight power loops, and the silkscreen was cleaned up (see PCB).

Order prep (18 Oct): F1 is now a 1206 chip fuse, 7 A / 63 V (Littelfuse 0458007.DR, LCSC C206980; second source Schurter 3413.0326.11). L2 and L3 now use the common 5040 shielded footprint: SWPA5040S470MT, 47 uH, 1.1 A (LCSC C86617; second sources Bourns SRN5040-470M, Taiyo Yuden NRS5040T470MMGK). EN_AUX was re-routed off the path between the L2 pads, and the VIN track at L3 now runs centred between the pads (0.4 mm each side). Every assembled part now carries an LCSC number checked against JLCPCB stock; the assembly upload files are DCDC_JLC_BOM.csv and DCDC_JLC_CPL.csv (217 parts). U1 (own sample), T1, J1-J5, J7, JP1 and SW1 are hand-soldered.

Rev E changes: The isolated 3.3 V from the isolator is gone. A small flyback (UC3843B, PA3855.006NLT EP13 transformer, TL431 + EL357N optocoupler feedback) makes an isolated +10 V for the secondary side from VIN, and the secondary rails now come from it instead of VBUS. U12 is now a CA-IS3742LW (data only). VREF+ is back on +3V3A (3.3 V full scale), and the REF3030 (U19) is removed. The PCB was re-placed and re-routed for a 1 oz outer / 0.5 oz inner stack: power on F.Cu pours, In1 as solid ground planes, gate loops routed as tight pairs. Schematic is 10 sheets (new sheet: Isolated aux flyback); 148 nets, 284 parts, 0 netlist errors.

Topology: LLC DCX + secondary 4SBB

An LLC alone cannot regulate this range. Frequency control gives roughly 0.7-1.5x gain, but 12-48 V in to 1.5-48 V out needs about 128:1. So the LLC runs at its resonant frequency as a fixed-ratio isolation stage (DCX), and the 4SBB does all the regulation on the secondary.

1. Primary: VIN -> half-bridge (Q5/Q6, NCEP0160AG) -> split resonant capacitors (2 x 300 nF C0G) -> PQ2620 primary. Lr is the transformer leakage, 1.0 uH.
2. Barrier: T1 (Np 4 : Ns 8+8) and U12 (CA-IS3642LVW). U12 carries the LLC gate signals to the primary and powers the secondary MCU at 3.3 V from the primary.
3. Secondary: centre-tapped synchronous rectifier (Q7/Q8, BSC160N15NS5, 150 V) -> isolated bus VBUS about equal to VIN (12-48 V) -> the 4SBB from the earlier design, unchanged.

Running at resonance gives ZVS on the primary and near-zero-current switching on the SRs at every input voltage, so the LLC is very efficient except at 12 V in, where its currents peak.

Start-up sequence:

1. VIN rises and the primary aux (U13 + U14) makes 10 V and 5 V. The flyback (U20, T2) starts and makes the isolated +10 V; U4 and U5 make 5 V and 3.3 V, and the MCU boots. CAN and the drivers are already powered.
2. The MCU starts the LLC with a duty and frequency soft-start: about 2.5 x fr and small duty, ramping to fr and 50% over about 10 ms. The SRs stay off, so the body diodes rectify.
3. VBUS rises. It now feeds only the 4SBB power stage, not the aux rails.
4. The MCU enables the SRs (SR_EN), then starts the 4SBB.

LLC sizing

The LLC runs at about 205 kHz with Cr = 600 nF, Lr = 1.0 uH (leakage) and Lm = 50 uH. ZVS needs Lm at or below 94 uH with 150 ns dead time, so 50 uH has about 1.9x margin at every input voltage. Peak flux is 63 mT at 48 V in, so core loss stays at or below 0.2 W.

VIN	Bus power	Primary rms	SR rms (each)	LLC loss	LLC efficiency	4SBB loss	Total loss	Total efficiency
12 V	53.6 W	10.1 A	3.6 A	3.9 W	93.2%	3.6 W	7.5 W	86.9%
18 V	53.1 W	6.7 A	2.4 A	2.0 W	96.3%	3.1 W	5.2 W	90.7%
24 V	53.2 W	5.0 A	1.8 A	1.4 W	97.4%	3.2 W	4.6 W	91.5%
36 V	53.4 W	3.4 A	1.2 A	1.0 W	98.1%	3.4 W	4.5 W	91.8%
48 V	53.7 W	2.6 A	0.9 A	1.0 W	98.1%	3.7 W	4.7 W	91.4%

Each row uses the 4SBB operating point that draws the most bus power at that input. SR drain voltage reaches 95 V at 48 V in (2 x VBUS), which is why the SRs are 150 V parts. Resonant capacitor voltage peaks at 28 V, inside the 100 V C0G rating. At 12 V in the primary FETs dissipate 0.6 W each and the transformer about 1.1 W, so give the primary side the same copper and via treatment as the 4SBB.

Transformer specification (custom, for a winding house): PQ2620 core in N97/PC95 or equivalent. Primary 4 turns rated for 10 A rms: 0.15 x 8 mm copper foil (litz 0.1 x 200 does not fit the 3.3 mm radial build of the vendor bobbin together with the secondary; litz 0.1 x 100 just fits). Secondary 8 + 8 turns centre-tapped in triple-insulated wire, 3.6 A rms per half. Lm 50 uH +/-10% (gapped), leakage 1.0 uH +/-20%, hipot 3 kVrms primary-secondary for 60 s. Leakage tolerance moves fr by about +/-10%, so firmware should trim the switching frequency to the measured VBUS/VIN ratio or to minimum primary current.

4SBB regulator sizing (secondary, input = VBUS)

The power stage runs at 150 kHz with L = 22 uH. Peak inductor current is 6.2 A and RMS is 5.05 A, both at 48 V in / 10 V / 5 A. Going from 200 kHz / 15 uH to 150 kHz / 22 uH cut the hottest FET loss from 1.69 W to 1.38 W.

FET choice follows what each switch does. Q1 and Q4 hard-switch, so they use the low-Qgd NCEP0140AG (100 V, 23 mOhm max, Qgd 3.8 nC). Q2 and Q3 carry the full 5 A in buck mode as synchronous rectifier and static-on switch, so they use the low-Rds NCEP0160AG (100 V, 8.5 mOhm max).

Operating point	Mode	Efficiency	Total loss	Q1	Q2	Q3	Q4	Inductor
12 V -> 5 V, 5 A	buck	89.5%	2.93 W	0.57 W	0.21 W	0.30 W	-	0.69 W
12 V -> 12 V,	transitio	93.5%	3.46	0.88 W	0.08 W	0.28 W	0.34 W	0.66 W

Operating point	Mode	Efficiency	Total loss	Q1	Q2	Q3	Q4	Inductor
4.2 A	n		W					
12 V -> 48 V, 1.04 A	boost	93.3%	3.62 W	0.60 W	-	0.09 W	1.38 W	0.62 W
24 V -> 24 V, 2.1 A	transition	95.5%	2.37 W	0.55 W	0.03 W	0.08 W	0.41 W	0.22 W
48 V -> 1.5 V, 5 A	buck	68.2%	3.50 W	0.99 W	0.33 W	0.30 W	-	0.68 W
48 V -> 10 V, 5 A	buck	93.2%	3.67 W	1.14 W	0.28 W	0.30 W	-	0.76 W
48 V -> 48 V, 1.04 A	transition	94.7%	2.79 W	0.80 W	0.01 W	0.03 W	0.76 W	0.16 W

Total loss includes 0.7 W housekeeping (MCU, CAN, aux supplies), which is why 1.5 V / 7.5 W shows 68%. Model assumptions: Rds at 1.4x the 25 C max, 8 ns V-I overlap per hard edge, 30 ns dead time, Qrr 50-80 nC (not listed for NCEP0160AG, so assumed), inductor DCR 25 mOhm.

Thermal: with 1.4 W in a DFN5x6 on 2 oz copper with a via array to inner planes (about 35-40 C/W), junction rise is about 50-55 C, so Tj is about 85 C at 30 C ambient. Minimum buck on-time is 234 ns at 48 V -> 1.5 V, well above the HRTIM's 184 ps resolution and the UCC27211's propagation delay.

Inductor: PSPMAA1770-220M-CGP, 22 uH, Isat 23 A, Irms 12 A, DCR 25 mOhm, 17 x 17 mm. Bourns SRP1770TA-220M is a drop-in alternate on the same footprint.

Control, sensing and MCU pin map

HRTIM Timer A drives the input leg and Timer E drives the output leg, both synchronised to the master timer. Each timer outputs a complementary pair with hardware dead time into one UCC27211. PA11/PA12 are kept for FDCAN1, which is why Timer B is not used. Timer C runs the LLC through the isolator, and Timer D runs the SRs, phase-locked to Timer C with about 30 ns lead to cover isolator and driver delay. In the 4SBB table below, VIN means VBUS, which tracks VIN within about 2%.

Inductor current is sensed by one 4 mOhm resistor in the common source return of Q2 and Q4 (the LM5175 scheme). It reads positive in boost and negative in buck, so U7 (INA181A1, gain 20) is offset to half of VREF+ (1.65 V). Output current for the CC loop uses a 10 mOhm low-side shunt and U8 (INA181A2, gain 50). VREF+ is tied to +3V3A again in Rev E (REF3030 removed) with 1 uF + 100 nF, so all ADC, DAC and comparator full scales are 3.3 V; the U7 offset is 1.65 V.

Pin	Port	Function	Net
42 / 43	PA8 / PA9	HRTIM1_CHA1 / CHA2 -> Q1 / Q2	PWM_A1 / PWM_A2

Pin	Port	Function	Net
40 / 41	PC8 / PC9	HRTIM1_CHE1 / CHE2 -> Q3 / Q4	PWM_E1 / PWM_E2
45 / 46	PA11 / PA12	FDCAN1_RX / TX	CAN_RXD / CAN_TXD
12	PA0	ADC1_IN1, VBUS about equal to VIN (100k / 5.1k, FS 61.8 V)	VBUS_SNS
33	PB11	ADC1_IN14 + COMP6_INP, VOUT + fast OVP	VOUT_SNS
13	PA1	ADC1_IN2, output current, 0.5 V/A	IOUT_SNS
25	PB1	ADC1_IN12 + COMP1_INP, positive OCP (boost)	IL_SNS
24	PB0	COMP4_INP, negative OCP (buck), polarity inverted	IL_SNS
8-11, 22, 23	PC0-PC5	Address bits 0-5	ADDR0-ADDR5
49 / 50 / 56	PA13 / PA14 / PB3	SWDIO / SWCLK / SWO	J6
19 / 59 / 60	PA5 / PB6 / PB7	DAC1_OUT2 test output / test GPIO 1 / test GPIO 2 (J6 pads 3-5)	TP_DAC / TP_GPIO1 / TP_GPIO2
62 / 52	PB9 / PC10	RUN / FAULT LEDs, active low	LED_RUN / LED_FLT
34 / 35	PB12 / PB13	HRTIM1_CHC1 / CHC2 -> U12 -> U10 -> Q5 / Q6 (LLC)	LLC_A / LLC_B
36 / 37	PB14 / PB15	HRTIM1_CHD1 / CHD2 -> U11 -> Q7 / Q8 (SR)	SR1_PWM / SR2_PWM
53	PC11	SR driver enable, active high	SR_EN
20	PA6	ADC2_IN3, NTC beside Q2 / RS1 (Rev D: moved off PA4)	NTC_SNS
5 / 6	PF0 / PF1	16 MHz HSE (Rev D), PLL M4 N85 R2 -> 170 MHz	OSC_IN / OSC_OUT
51	PA15	HRTIM1_FLT2 (AF13), input OVP, active high (Rev D)	VIN_OVP
28	VREF+	ADC / DAC / COMP reference input = +3V3A (3.3 V, FB1-filtered); VREFBUF disabled	VREF_ADC
14	PA2	TIM15_CH1 (AF9), PWM input: isolated VIN	VIN_PWM

Pin	Port	Function	Net
		monitor, 10k pull-down R93 (Rev D)	
18	PA4	Reserved for DAC1_OUT1, 10k pull-down	SP_PA4
2-4, 17, 21, 26, 30, 38, 39, 44, 54, 55, 57, 58	PC13-15, PA3, PA7, PB2, PB10, PC6, PC7, PA10, PC12, PD2, PB4, PB5	Unused: 10k to GND each (R90-R105, except R93, which is now the pull-down for PA2 = VIN_PWM, TIM15_CH1)	SP_<port>

Every pin, with all alternate functions (AF0-AF15) copied from ST's pin database and a column for what this design uses, is in **pin-map.xlsx** (01_hardware/docs).

Comparator thresholds come from DAC3, and COMP1/4/6 outputs feed HRTIM fault inputs internally for cycle-by-cycle shutdown with no external parts. The INA181 bandwidth (350 kHz) is fine for averaged control and OCP but too slow for true peak-current-mode control, so plan on voltage-mode or average-current-mode loops.

Firmware must handle these board-specific items:

- Bootstrap refresh: Q1 is static-on in boost and Q3 in buck. Cap their duty at about 98% or insert a short low-side pulse every few cycles so the bootstrap capacitors stay charged.
- Output voltage correction: the VOUT divider is referenced to board GND, so VOUT = V_divider + IOUT x 10 mOhm (up to 50 mV at 5 A).
- Current-sense zero: U7's REF is a 1k / 1k divider from VREF, which leaves a small gain and offset error. Measure IL_SNS with all FETs off at start-up and subtract it.
- ADC reference: VREF+ = +3V3A (U5 TLV75533, 1 %, through FB1). Leave the internal VREFBUF disabled, as it is after reset (VREFBUF_CSR: ENVR = 0, HIZ = 1). Scale every ADC, DAC3 and comparator value to 3.3 V full scale, and correct the gain at run time with VREFINT against VREFINT_CAL (actual VDDA = 3.0 V × VREFINT_CAL / VREFINT_DATA).
- VIN monitor: TIM15 in PWM-input mode on PA2, VIN = 50.47 V × (t_high − t_reset) / (period − t_reset) − 0.20 V, t_reset about 20 µs, average 8–16 periods, two-point calibration stored in flash.
- Input OVP: configure HRTIM1_FLT2 on PA15 active-high so VIN_OVP shuts all HRTIM outputs in hardware.

Full-scale and threshold summary

Rev E: every ADC channel, DAC3 and the comparators use VREF+ = +3V3A = 3.3 V. +3V3A is the MCU's 3.3 V rail (U5 TLV75533, ±1 %) through FB1 600R with 1 µF + 100 nF at VREF+ (C36/C37) and VDDA (C28/C29). The external REF3030 (U19) is removed. The IL zero (R25/R26) and the NTC pull-up (R29) hang on +3V3A, so those two channels stay ratiometric. Firmware keeps VREFBUF off (VREFBUF_CSR: ENVR = 0, HIZ = 1).

Signal	MCU input	Conversion	Full scale	1 LSB	Normal use
VBUS (about VIN)	PA0, ADC1_IN1	100k / 5.1k, 0.1 %: ÷20.61	0–68.0 V	16.6 mV	12–48 V = 18–71 % of FS
VOUT	PB11, ADC1_IN14 + COMP6	100k / 5.1k, 0.1 %: ÷20.61	0–68.0 V	16.6 mV	1.5–48 V; add IOUT × 10 mΩ
Inductor current IL	PB1 ADC1_IN12 + COMP1; PB0 COMP4	4 mΩ × 20 = 80 mV/A, zero at VREF/2 = 1.65 V	±20.6 A (about ±19.5 A usable, INA181 output swing)	10.1 mA	peak 6.2 A = 30 % of FS
Output current IOUT	PA1, ADC1_IN2	10 mΩ × 50 = 0.5 V/A	0–6.6 A	1.6 mA	5 A = 2.5 V (76 % of FS)
Temperature NTC	PA6, ADC2_IN3	10k pull-up from +3V3A, 10k B3950 to GND	−40 °C (3.22 V) to 125 °C (0.11 V)	0.02 °C at 25 °C, 0.14 °C at 100 °C	25 °C = 1.65 V, 100 °C = 0.21 V
VIN (isolated monitor)	PA2, TIM15_CH1	VIN = 50.47 V × x − 0.20 V (see VIN monitor)	0–50.5 V; readable 9.7–50.3 V	about 1 mV (51,700 counts)	12–48 V
VIN over-voltage	PA15, HRTIM1_FLT2	Primary comparator, digital	Trips at 50.26 V rising, clears at 49.83 V (±1.3 %)	—	Below the 51 V TVS stand-off
Fast fault thresholds	DAC3 into COMP1 / 4 / 6	12-bit, 0–3.3 V	IL ±20.6 A, VOUT 0–68.0 V	10.1 mA, 16.6 mV	Set per mode by firmware

Error budget, untrimmed: VBUS and VOUT about ±1.2 % (U5 ±1 %, 0.1 % dividers, ADC). Correcting the gain with VREFINT at run time brings this to about ±0.5 %, and a one-point calibration of VBUS/VOUT at room temperature to about ±0.3 %. IL and the NTC are ratiometric, so the reference error cancels. IOUT is about ±1.5 % (INA181 gain ±1 %, 1 % shunt) plus offset, so zero IL and IOUT at start-up with the power stage off. VIN is about ±1.5 %, or about ±0.5 % after a two-point calibration (separate TL431 reference on the primary side).

Input over-voltage protection (Rev D)

A comparator on the primary side watches VIN and reports through the spare forward channel of the isolator U12 (VI1 -> VO1), so no extra barrier part is needed.

Part	Value	Job
R81 / R82	191k (1206) / 10k, 0.1 %	VIN divider, 1 nF C0G filter (C79)

Part	Value	Job
U16	TL431AIDBZR	2.495 V reference, ±1 % (A grade; TL431BIDBZR is a 0.5 % drop-in), 1k bias from +5V_P (R85)
U15	LMV331IDBVR	Open-drain comparator, + = divider, - = reference
R83 / R84	10k / 2.2M	Pull-up to +5V_P / hysteresis

Trip point = 50.26 V rising and 49.83 V falling (0.43 V hysteresis). This is 2.495 V × (1 + 191k / (10k ‖ 2.2M)), less about 0.2 V from the VIN-monitor feedback R110. Tolerance is about ±1.3 % (TL431A ±1 %, comparator offset), so 49.6–50.9 V, still below the 51 V TVS stand-off. Above it OVP_P goes high, U12 VO1 drives VIN_OVP high and PA15 (HRTIM1_FLT2, active high) shuts every HRTIM output in hardware. The 51 V TVS still clamps fast transients; the comparator stops switching for sustained over-voltage.

Test points (Rev D) are 1.5 mm pads on quiet nodes only: rails VIN, +10V_P, +5V_P, VBUS, +10V, +5V, +3V3, +3V3A, VOUT, the grounds GNDP / GND, and the filtered IL_SNS, IOUT_SNS and VIN_OVP. No test pad sits on a switch node, gate or high-impedance divider, so they add no antenna. TP1-TP4 are primary-side and float relative to earth.

Input voltage monitor (PWM over isolation)

VIN is read for monitoring only, not for control. The reading crosses the isolation barrier on the spare isolator channel, so the circuit needs no isolated amplifier. It reuses the OVP divider (R81/R82, now 0.1 %) and the TL431 reference (U16) that are already on the primary side.

Block	Parts	What it does
Ramp	Q9 MMBT3906, R86 56k, R87 10k, R88 15k, C82 47 nF C0G	PNP current source (~24 µA) charges C82 linearly from 0 to VREF_P in about 5 ms (~200 Hz)
Reset	U17 LMV331, R106 1k, D7 1N4148W, C81 220 pF C0G, R107 100k, Q10 2N7002, R89 22 Ω	When RAMP passes VREF_P, U17 releases and charges C81 through D7 in about 0.2 µs. R107 then holds Q10 on for about 15–30 µs, which empties C82 to under 1 mV
PWM comparator	U18 LMV331, R109 1k, R110 4.7M, R108 10k	IN+ = OVP_DIV, IN− = RAMP (through R109). VINPWM_P is high while RAMP < OVP_DIV, and during the reset. R110 feeds about 10 mV of hysteresis back into the low-impedance divider, never into the ramp
Isolation	U12 channel 2 (pin 4 → pin 13)	VINPWM_P on the primary side becomes VIN_PWM on the secondary side
MCU	PA2 (pin 14), R93 10k	TIM15_CH1 in PWM-input mode measures the

Block	Parts	What it does
	pull-down	period and the high time

Firmware formula. Run TIM15 in PWM-input mode on PA2 (prescaler 16, giving about 51,700 counts per 4.9 ms period). With x = (t_high − t_reset) / (period − t_reset) and t_reset about 20 µs: VIN = 50.47 V × x − 0.20 V. The 50.47 V includes R84 (OVP_P low in normal running) and R110; the −0.20 V is the R110 feedback. Full scale is 50.5 V, and readings are valid from about 9.7 V (primary aux start) up to the OVP trip. Average 8–16 periods (one reading every 40–80 ms).

Accuracy. A behavioural simulation of the final circuit gives the formula within 0.02 % from 12 to 48 V, so the ramp is linear. Untrimmed, expect about ±1.5 %: TL431A ±1 %, comparator offsets up to ±7 mV, and the spread of t_reset (15–30 µs, from the 2N7002 threshold). Use a **two-point calibration** (for example 24 V and 48 V, gain and offset stored in flash), because the t_reset spread acts as an offset. After calibration expect about ±0.5 % over temperature, limited by TL431 drift. Isolator delay skew (a few ns) is negligible against a 5 ms period.

Review fix (18 Oct). The first version took the U18 hysteresis from the comparator output back onto the ramp (R109 10k / R110 1M). That 1 MΩ path drew up to 10 % of the 24 µA ramp current and reversed direction when U18 switched. The ramp bent, and simulation showed reading errors of +12.5 % at 12 V and +1.5 % at 48 V, with 3.6 % left even after a two-point calibration. The hysteresis now goes into the divider: U18 inputs swapped, R110 4.7 M to OVP_DIV, R109 1k. The reset pulse was also hardened: C81 1 nF → 220 pF and R107 22k → 100k. The gate now charges fully before U17 lets go, so C82 always discharges to near 0 V.

On the PCB, the block sits on the primary side below U15/U16 (x 40–64 mm, y 85–97 mm). VIN_PWM leaves U12 pin 13 through a via and runs on inner and bottom layers to R93/PA2. Clearance from primary copper to secondary copper is unchanged.

Auxiliary supplies, isolated CAN, addressing

There are now two auxiliary chains, one on each side of the barrier.

Side	Chain (Rev E)	Feeds	Load
Primary (GNDP)	VIN -> U13 LMR16006X -> +10V_P -> U14 AMS1117-5.0 -> +5V_P	U10 LLC driver, U20 UC3843B, U12 side A, OVP / VIN monitor	42 mA at 10 V (U13 rated 600 mA)
Barrier	U20 flyback, T2 PA3855.006NLT (2250 Vdc hi-pot), feedback through U21 EL357N	Isolated +10 V on GND	135 mA = 1.35 W (budget), 2 A primary peak limit
Secondary (GND)	+10 V (flyback) -> U4 LMR16006X -> +5V -> U5 TLV75533 -> +3V3 -> FB1 -> +3V3A	U2/U3 4SBB drivers, U11 SR driver (10 V); CAN VCC (5 V); MCU, U12 side B,	5 V 175 mA, 3.3 V 55 mA

Side	Chain (Rev E)	Feeds	Load
		sensing (3.3 V)	

Flyback (Rev E). U20 UC3843B runs from +10V_P at 172–180 kHz in DCM, switching Q11 FDN86246 (150 V) into T2 (Np:Ns 1.33:1, Lp 21 µH, two secondaries in parallel, aux winding unused). R114 0.5 Ω limits the primary peak to 2 A (Bpk 1170 G, under the 2700 G limit). The RCD clamp D8 / R116 4.7k (1206) / C94 4.7 nF holds Vds under 85 V at 50 V in. D9 SS110 rectifies into C96/C97.

Regulation is on the secondary: TL431 U22 with R119 30k / R120 10k sets +10 V = 9.98 V (9.68–10.28 V worst case). It drives the EL357N LED through R117 1k5 (R118 1k bias gives the TL431 at least 1 mA), and the opto collector pulls U20 COMP down (U20 FB grounded). Type-2 compensation R121 4.7k / C99 100 nF puts the zero at 340 Hz; check the crossover with a 50–100 % load step on the bench. Ripple is about 145 mV p-p at 3 W and 65 mV at the normal load. Full numbers are in calc/flyback_aux.txt.

Both LMR16006 bucks use a 47 uH 5040 shielded inductor (L2, L3, 1.1 A saturation, 354 mOhm). U13 (primary) still makes +10V_P from VIN and now carries about 42 mA, including the flyback controller. U4 (secondary) is now a 10 V to 5 V buck fed by the flyback (R18 56k / R19 10k for 5.05 V, EN R16 56k / R17 15k for about 5.8 V start) and carries about 175 mA. Both inductors stay far below saturation, with copper loss under 10 mW each.

Rev E: every secondary rail (MCU 3.3 V, 5 V for CAN, 10 V for the drivers) comes from the flyback as soon as VIN is present, so the MCU, CAN and the 4SBB drivers are all up before the LLC starts. The LLC no longer has to start blind at power-up, and the MCU can check VIN (OVP / VIN monitor) first. U13 starts at about 9.7 V through its EN divider (470k / 68k), so the whole board starts at VIN of about 10 V.

CAN uses the CA-IS3062VW, which has its own isolated DC-DC, so no separate isolated module is needed. VCC is 5 V and VCCL is 3.3 V, so TXD/RXD connect directly to the MCU. The bus side has split termination (2 x 60.4 Ohm + 4.7 nF) enabled by JP1, a 2x2 header that breaks both legs (pins 1-2 CANH to R30, pins 3-4 R31 to CANL; fit both shunts on an end-of-bus node). With no shunts the termination is fully off both lines, so CANH no longer sees the 60 Ohm + 4.7 nF load that the earlier single-jumper layout left on it, an NUP2105L TVS, and two 3-pin connectors for daisy-chaining. A 2.2 nF / 2 kV Y-cap across the barrier is placed but marked DNP.

Address bits are read on PC0-PC5 with 10k pull-ups. DIP switch ON pulls a bit to 0. Header J5 is wired in parallel through 1k series resistors, so a harness that grounds a pin also sets that bit to 0.

The 6-pin header has no ground pin of its own. It relies on the harness sharing ground through the power connector. If the harness is not ground-referenced, J5 needs a 7th pin.

Connectors, programming and mounting

Ref	Function	Part	Pins
J1	Input 12-48 V (primary)	KF2EDGR-5.08-2P pluggable male header, 5.08 mm, 12 A	1 VIN, 2 GNDP
J2	Output 1.5-48 V	KF2EDGR-5.08-2P pluggable male	1 VOUT, 2 VOUT- (after the 10

Ref	Function	Part	Pins
		header, 5.08 mm, 12 A	mOhm shunt)
J3, J4	CAN in / out (daisy chain)	2.54 mm 1x3 male pin header	1 CANH, 2 CANL, 3 GND_ISO
J6	Programming pads (pogo fixture)	8 round SMD pads, 2.0 mm, 2.54 mm pitch, pad 1 square	1 SWDIO, 2 SWCLK, 3 Test_pin_DAC (PA5), 4 Test_pin_GPIO1 (PB6), 5 Test_pin_GPIO2 (PB7), 6 NRST, 7 VCC_3.3, 8 GND
J7	ARM Cortex Debug 10-pin	2x5 1.27 mm SMD header (FTSH-105 or equivalent)	1 VTref 3.3 V, 2 SWDIO, 3/5/9 GND, 4 SWCLK, 6 SWO, 7 key, 8 NC, 10 nRESET
H1-H4	M3 standoffs	3.2 mm non-plated holes, 4 mm from each corner	-

The old 8-pin SWD/UART header is gone, so USART2 (PA2/PA3) is no longer brought out; PA3 is free, and PA2 now reads the VIN monitor PWM (TIM15_CH1).

The schematic is now 10 sheets: a top sheet plus one A3 sheet per block (LLC stage, primary aux and isolator, isolated aux flyback, 4SBB power and drivers, secondary aux, MCU, sensing, CAN, IO). Parts inside a sheet are joined by wires, rails use power symbols, and only signals that go to another sheet carry a global label.

Rev E2 debug-port series resistors: SWDIO, SWCLK, TP_DAC, TP_GPIO1, TP_GPIO2 and NRST reach the J6 pads and the J7 SWD header through 47.5 Ω 0402 resistors (R123–R128), and J6 pin 7 / J7 VTref take +3V3 through a 2 Ω 0603 (R129, net PROG_3V3). They damp ringing on the probe cable and limit the current into a pin if a pad is shorted or a probe is plugged in wrongly; 47.5 Ω is far below anything that matters for SWD at the usual 1–4 MHz clock. Each resistor sits on the MCU side of the point where a trace splits to J6 and J7, so both connectors go through it.

BOM and sourcing

The estimated BOM is about $37.14 per board at 100 pcs: 227 placed parts on 90 lines, excluding PCB and assembly. The isolation stage (LLC, isolator, primary aux, OVP and VIN monitor) adds about $13.04, which is 54% over the $24.19 non-isolated design (which now includes the $0.50 REF3030 ADC reference). That misses the 20% premium target.

Rev E has not been re-costed yet. It adds the flyback (UC3843B, PA3855.006NLT, FDN86246, EL357N, TL431, about 20 passives) and swaps the CA-IS3642LVW for the cheaper CA-IS3742LW. It also drops the REF3030 and the secondary VBUS-fed 10 V stage. The JLC BOM has 82 lines (duplicate LCSC numbers merged), and every assembled part has an LCSC number.

Isolation-stage part	Qty	Cost @100
CA-IS3642LVW isolator + isolated 3.3 V (C7422564)	1	$2.51
PQ2620 transformer, custom wound (estimate)	1	about $2.50
UCC27211DDAR primary driver (C141980)	1	$1.61
BSC160N15NS5 SR FET (C459172)	2	$1.50
NCEP0160AG primary FET (C503012)	2	$1.47
2.2 uF 100 V X7R 1210	4	$1.45
100 nF 100 V C0G 1206, resonant caps (estimate)	6	$0.48
LMR16006X + AMS1117-5.0 + passives, primary aux	-	about $0.80
UCC27524DR SR driver (C465729)	1	$0.21

The nearest realistic cost-downs are about $2.30, which would bring the premium to about 45%:

- Generic 100 V X7R 1210 capacitors instead of the Samsung part (about -$1.00).
- A lower-cost half-bridge driver in place of UCC27211 on the primary (about -$1.30).

Getting to 20% would take a different isolation approach, for example a lower-power isolated stage or a cheaper digital isolator plus a discrete isolated supply. The 20% target may have been set against a different baseline, so it's worth confirming which one applies.

The remaining key parts, below, are unchanged from the non-isolated design.

Part	LCSC	Price @100	Stock (2026-10-18, JLCPCB)
STM32G474RBT6	C529413	$7.24	Out of stock. First build uses your own samples (hand-soldered). Pin-compatible STM32G474RET6 C521608: 354
CA-IS3062VW	C5338808	$2.51	In stock
CA-IS3642LVW	C7422564	$2.51	2, rest on JLC pre-order. CA-IS3642LW (no isolated power) and HVW (outputs default high) are not substitutes
UCC27211DDAR x3	C141980	$1.61	In stock. Second source SLM27211CA-DG C967931 is out of stock
PSPMAA1770-220M-CGP	C49308184	about $1.00	In stock. Alternate: Bourns SRP1770TA-220M
NCEP0160AG x4	C503012	$0.74	In stock
NCEP0140AG x2	C503009	about $0.40	In stock

Part	LCSC	Price @100	Stock (2026-10-18, JLCPCB)
FS32X225K101EGG 2.2 uF 100 V X7R 1210 x13	C153036	about $0.36	69 (enough for 5 boards). Samsung CL32B225KCJSNNE C55151 is out of stock; any 100 V X7R 1210 works
C1206N104J101T 100 nF 100 V C0G 1206 x6 (CR1-CR6)	C2940749	about $0.08	892
LMR16006XDDCR	C87080	$0.36	In stock
SWPA5040S470MT 47 uH 1.1 A x2 (L2, L3)	C86617	about $0.12	24,085. Same 5040 footprint: SRN5040-470M, NRS5040T470MMGK, NRSC5040-470M
0458007.DR 7 A 63 V 1206 fuse (F1)	C206980	about $0.45	376. Same 1206 footprint: Schurter 3413.0326.11
INA181A1IDBVR / INA181A2IDBVR	C2058943 / C2058784	$0.24 each	In stock
REF3030AIDBZR (ADC reference, removed in Rev E)	C38423	about $0.50 (estimate)	In stock

Every assembled part has an LCSC number checked in the JLCPCB parts library on 18 Oct (package, rating and stock). Small resistors and caps are JLC basic parts where one fits. The 0.1 % dividers are thin film, 25 ppm/°C, and CR1-CR6 are C0G. Two parts need attention: U12 has only 2 in stock, so JLC pre-orders the rest, and C82 is X7R because JLC stocks no 47 nF C0G 0805. The VIN monitor is ratiometric and calibrated, so the X7R C82 is acceptable.

PCB (Rev D) and layout review

Rev E board: 160 x 105 mm, 4 layers, 1 oz outer / 0.5 oz inner, 284 footprints, 1,463 track segments, 772 vias and 44 zones. KiCad DRC with the JLCPCB rules (DCDC_LLC_4SBB_G474.kicad_dru) reports 0 clearance errors, 0 unconnected items and 0 silkscreen-overlap or text-size errors. The remaining DRC lines are 199 "footprint differs from library" notices, 16 courtyard overlaps that were checked and accepted (SMD parts inside a transformer or MCU courtyard but outside its body, and the DNP R122/C98 pair), and 4 footprint silk marks that JLC clips at the pads. 53 references on the densest small parts are hidden on the silkscreen; every reference stays on the assembly (fab) layer.

Rev E layout and copper

The board was laid out again for 1 oz outer copper. Power runs on F.Cu pours, one net per pour. In1 is solid ground (GNDP / GND / GND_ISO) with 4 mm keep-out bands at the barrier. B.Cu carries the VBUS link, the VOUT strip and a copy of the PGND bar. In2 and B.Cu carry the signals under a ground flood.

Copper loss in the main current paths at the 50 W / 12 V-in corner totals about 0.5 W, with no path above 90 mW (calc/copper_check.txt).

Path	Current (A rms)	Narrowest (mm, F.Cu)	Layers	Loss (mW)
LLC half-bridge -> T1 (HB_SW)	10.1	6.4	F	28
T1 -> resonant caps (CR_MID)	10.1	5.0	F	90
VIN: F1 -> C60..C61 -> Q5	4.6	7.2	F	44
SR side -> 4SBB input (VBUS)	4.5	1.2	F + B link	26
Q1/Q2 -> L1 (SW1)	5.2	6.8	F	25
Q2 sources -> RS1 (PGND)	5.0	1.8	F + B copy	27
Q3 -> output caps -> J2 (VOUT)	5.0	3.4	F + B	69
RSH1 -> RS1 -> SR return (GND)	5.0	1.3 (In1)	In1 + B + F	50

IPC-2221 (outer layer, 1 oz, 20 °C rise) needs about 1.8 mm for 5 A and 4.7 mm for 10 A. The two narrower necks are short and have a parallel layer.

Other layout choices in Rev E:

- Every gate loop is a tight pair (HO + its HS return, 0.4 mm, 0.4 mm gap) on B.Cu or In2. The bootstrap cap and gate resistor sit beside the driver so HO leaves straight through the resistor (U10/R62/C67, U2/R4/C15).
- Power via fields (VBUS, VOUT, PGND top-to-bottom) use 0.6/0.3 mm vias at 1.7 mm pitch. That leaves a 0.5 mm ground web between the antipads on In1, so the plane under the power stage has no slot.
- Each small R/C ground pad in the MCU area has its own via, and the MCU GND pins drop to vias inside the pin ring. C1/C5 returns have vias right beside the pads.
- The router kept 0.45 mm off the board edge and 0.28 mm from PTH pads, and treated the power pours as obstacles.
- No copper crosses the 4 mm isolation bands. Only T1, T2, U12 and U21 bridge them.

Layer	Use
Top	Parts. Hot-loop copper is pre-routed at 1.0-2.5 mm and locked, with local power pours around each loop and a ground flood per domain
Inner 1	Solid split planes GNDP / GND / GND_ISO directly under every power loop
Inner 2	Signals, ground flood in the free space
Bottom	Signals, VBUS bar from the T1 centre tap to the 4SBB, VOUT_N bar, ground flood

Power path

- **LLC primary.** C61-C63 straddle the Q5 drain and the Q6 source, so the half-bridge loop is about 9 x 8 mm over the GNDP plane. The HB_SW node is a small pour going straight into T1 pins 1-3. CR1-CR6 sit in one column against the T1 CR_MID pins.
- **Input.** J1 -> F1 -> D1 / C60 -> the C61-C63 bank on 2.5 mm copper plus a VIN pour.
- **Synchronous rectifier.** The Q7/Q8 drains face the T1 SR pins (about 4 mm). The VBUS ceramics C2-C4 sit between the T1 centre tap and the SR sources.
- **VBUS to the 4SBB.** A 2.5 mm bottom-layer bar plus four 0.8 mm vias runs from the T1 centre tap to C5/C6/Q1.
- **4SBB.** It follows the LM5175 layout: L1 on top, Q1/Q2 on the left, Q3/Q4 on the right. The low-side sources face each other, joined by a PGND bar into RS1. The input ceramics are at the Q1 drain and the output ceramics at the Q3 drain. The ground returns run on Inner 1 directly under each loop.
- **Output.** Q3 -> C7-C12 -> C11 -> J2 on 1.5-2.5 mm copper plus a VOUT pour. VOUT_N runs from J2 to RSH1 on a bottom-layer bar.

Noise and interference

- Switch nodes (HB_SW, SW1, SW2, SR_D1, SR_D2) are kept to small pours. They carry no test pads, and the Inner 1 ground shields every Inner 2 or bottom signal crossing under them.
- Kelvin sensing: U7 taps the inner edges of the RS1 pads and U8 taps RSH1. Neither amplifier input gets a plane via. The NTC now sits between Q2 and RS1, the hottest spot.
- MCU: U1 is rotated so that its HRTIM and sense pins face the power stage. A GND island under the LQFP with 16 vias to Inner 1 serves as the return for all VSS pins and decoupling. Every unused pin has its own 10k pull-down, fanned out so escape routes stay free.
- Isolation: 4 mm no-copper bands on all layers. Only T1, U12 and C66 bridge primary to secondary, and only U9 and C56 bridge to the CAN side.

Review before ordering

- **Gate-drive loops are longer than ideal.** Measured from driver pin to FET gate they run 14-42 mm: U2 -> Q1 is 42 mm, U3 -> Q3 31 mm, U11 -> Q8 30 mm, and U10 -> Q5/Q6 24-26 mm. Pulling U2, U3 and U11 closer to their FETs, with each gate resistor next to its gate pin, would cut ringing and EMI. This is the first thing I would hand-tune in EasyEDA.
- The crystal traces are 10-11 mm with two vias each. Keep them on the top layer next to Y1 if you touch that area.
- The long HRTIM, ADC and CAN lines (30-75 mm) run on Inner 2 / bottom over the ground plane. That is fine for these edge rates, but do not route them under L1 or the switch nodes when editing.
- T1 footprint takes either vendor bobbin: PQ2620 6+6 (25.4 mm rows, 0.70 mm pins, 29 x 27 mm base, 20.8 mm high) or PQ2625 6+6 (25.5 mm rows, 0.80 mm pins, 29.8 x 26.7 mm base, 25.3 mm high). Same 3.8 / 7.5 mm pin pattern; 1.2 mm holes; outline sized for the PQ2625. CR1-CR6 moved 0.7 mm left, Q7/Q8 0.68 mm right, T1 0.1 mm right to clear it.

- Silkscreen: all text is at least 1.0 mm high with a 0.15 mm stroke (JLCPCB minimum), clear of pads and other silk. C2–C4 (VBUS ceramics at the SR) and C67 (HBP bootstrap) are too crowded for a reference, so theirs are hidden and shown only on the assembly (Fab) layer.
- Optional: a milled slot in the primary/secondary band, between T1 and U12, if your standard needs more creepage.
- ADC reference: on the first board, check VREF+ with a scope while the ADC is converting. The REF30 datasheet asks for care with large low-ESR ceramics on its output; if VREF+ rings, put 1–2 Ω in series with C36.
- Rev E flyback bring-up: power the primary from a current-limited supply at 12 V, check +10 V (9.7–10.3 V) at no load and at 135 mA, look at Q11 Vds at 50 V in (should stay under 85 V), and step the load 50–100 % to check the loop settles without ringing.
- Rev E2 (20 Oct): R123–R129 were put into the existing debug traces, keeping the routing as it was. The Rev E DRC failures are fixed as well: 16 courtyard overlaps (at T1/CR1–CR6 and R64, at T2 with D8, C94, R116, C95 and the flyback FET Q11 and its shunt R114, plus C98/R122, C69/U11 and C34/U1) were cleared by moving those parts 0.05–2 mm and re-attaching their tracks, and the 4 silkscreen warnings went with them. Checks afterwards: DRC 0 errors, 0 warnings, 0 unconnected; schematic-to-PCB parity 291 parts, 796 pads, no mismatch.

Layout review and clean-up (18 Oct)

A full pass over noise, interference and loss. The table lists what was found and changed on the board; checks afterwards: DRC 0 errors, netlist parity 237 parts / 125 nets, primary-secondary and secondary-CAN copper gaps still 4.0 mm on every layer.

Finding	Effect	Change	Result
Output path C11 to J2 was one 1.0 mm track, 21 mm long	about 10 mOhm, 0.25 W and 50 mV drop at 5 A	VOUT pour along the path	about 3.5 mOhm, 0.09 W
CR1-CR6 only touched the pours at their pad edges after the PQ2625 move	10 A rms primary current through 1.0 mm tracks	VIN and GNDP pours over the left pads, CR_MID pour extended over the right pads	full-width connection on both sides
4SBB switch-node copper SW1/SW2 larger than needed	radiates and couples to chassis/heat sink	SW1/SW2 pours cut to L1 pad, FET pins/tab and the path between; driver-return traces on In2 narrowed	SW1 282 to 213 mm2, SW2 271 to 211 mm2
Q1 high-side gate loop: return trace far from the gate trace	ringing, slower and noisier switching	SW1 driver return re-routed on In2 0.8 mm beside G1	loop about 85 to about 15 mm2
Q3 high-side gate loop; G3 was 0.25 mm on B.Cu	same	G3 0.5 mm; SW2 return on B.Cu parallel to it from a via at the Q3	loop about 47 to about 18

Finding	Effect	Change	Result
		source pins	mm2
Q5 (LLC high side) gate loop	same, smaller	HB_SW return on B.Cu right beside the G5 run	loop about 17 to about 8 mm2
U8 sense lines tapped the GND pour and a load-carrying track instead of the RSH1 pads	small output-current error	both taps moved inside the RSH1 pads (true Kelvin)	U7 was already Kelvin
Router staircases (1400 segments under 0.45 mm)	extra corners, hard to edit	replaced by straight or 45-degree runs, clearance-checked	2906 to 1810 segments
Ground floods only loosely tied to the In1 planes	floods resonate, weaker return paths	320 stitching vias on a 5 mm grid where a ground net is poured on all four layers (GND 110, GNDP 194, GND_ISO 16)	427 to 747 vias in total

Left as is, with the reason:

- In1 planes are continuous: the longest antipad gap is 3.8 mm, a row of VBUS vias away from any return path.
- The SR loop (T1 secondary, Q7/Q8, C2-C4) and the LLC half-bridge loop are already compact.
- LLC gate drive to Q6 returns through the GNDP plane directly under it.

Recommended for Rev E (needs placement changes):

- Move U2 and U3 next to Q1/Q2 and Q3/Q4. They sit about 20 mm away; the pairing above limits the loop area but not the inductance of the length.
- The boost-mode hot loop runs from C7-C10 (x 143-153) back to RS1 (x 122). Move two of the 2.2 uF output MLCCs next to Q3/Q4, or add a 100 nF MLCC from VOUT to PGND right at Q4, so the high-frequency loop closes at the FETs.

Importing into EasyEDA Pro, and layout

Import the zip from the EasyEDA Pro start page, or through its Format Converter. EasyEDA's documentation lists KiCad 5.1 as supported and is unclear about newer versions, so the schematic is written in KiCad 6 format, the most widely accepted. The PCB file is KiCad 7 format because that is what the generator writes. (EasyEDA import docs, format converter)

1. Import DCDC_LLC_4SBB_G474_EasyEDA_import.zip. It carries its own library for the PQ2620 transformer and the programming pads (custom_fp).
2. If the PCB is rejected, import the schematic alone. Every part carries LCSC and MPN fields, so you can swap in EasyEDA library parts, which bring their own footprints.

3. Check the result against the schematic and PCB PDFs. The schematic is 9 wired sheets. The PCB is the routed Rev D board (see PCB).

To open the project in KiCad instead, install KiCad 7 or newer with its standard libraries (the default Windows installer includes them). The project's own sym-lib-table and fp-lib-table add the custom libraries. With the standard libraries present, DRC also stops reporting "footprint differs from library".

Open questions

- ☐ Is 48 V the absolute maximum input, or are there transients above it? The 51 V TVS and 63 V electrolytics assume 48 V maximum.
- ☐ Is reverse-polarity protection needed beyond fuse + TVS? An ideal-diode FET would add about $0.5 and 0.1-0.2 W.
- ☐ Does the address harness share ground with the power input? If not, J5 needs a ground pin.
- ☐ Which baseline does the 20% two-stage premium refer to? Against the non-isolated design the LLC stage adds about 55%.
- ☐ Which safety standard and pollution degree apply? That sets the final creepage and transformer construction for basic insulation.
- ☐ Is 87% efficiency and 7.5 W loss at 12 V in acceptable for passive cooling? If not, the primary could become a full bridge, roughly halving the primary conduction loss for about $2 more.
- ☒ PQ2620 bobbin row-to-row spacing: 25.4 mm from the vendor drawing (footprint updated).
- ☐ Footprints to confirm against the parts you buy: NCEP01x0AG DFN5x6 (placed as Infineon TDSON-8-1), the 22 uH / 23 A inductor (Bourns SRP1770), KF2EDGR-5.08 (Phoenix MSTBA 5.08), the UCC27211DDAR exposed pad, the SW1 DIP-6 part number, and the 1206 fuse F1. At 12 V in and full 50 W, F1 carries 4.8 A, 68 % of its 7 A rating against the 75 % (5.25 A) continuous limit at 25 C, so check its temperature in the 12 V full-load test.

Sources: component data from the LCSC/JLCPCB listings and datasheets linked above, and the CA-IS3062 datasheet, UCC27211 datasheet and INA181 datasheet. STM32G474 pin functions are from the STM32CubeMX pin database.